Probability & Statistics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Random sampling

What is the value of $16^{\frac{3}{4}}$?

2 [Foundational] • Target: Estimating proportions from samples

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?

3 [Foundational] • Target: Describing shapes of data distributions

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

4 [Foundational] • Target: The normal distribution

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

5 [Foundational] • Target: Interpreting and comparing z-scores

What are the center and radius of the circle given by the equation $(x - 2)^2 + (y + 6)^2 = 36$?

6 [Foundational] • Target: Normal distribution probabilities

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, a_3?

7 [Foundational] • Target: Random sampling

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Estimating proportions from samples

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of k?

9 [Intermediate] • Target: Describing shapes of data distributions

What is the center of the circle represented by the equation $(x - 3)^2 + (y + 5)^2 = 16$?

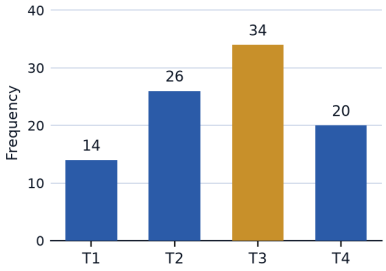
10 [Intermediate] • Target: The normal distribution

Solve: $\sqrt{2x + 5} - \sqrt{x + 2} = 1$.

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11 [Intermediate] • Target: Describing shapes of data distributions

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(5)$.

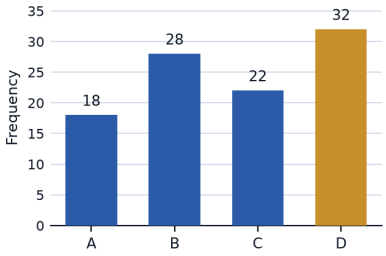


12 [Intermediate] • Target: Describing shapes of data distributions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

13 [Intermediate] • Target: The normal distribution

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(2)$.



14 [Intermediate] • Target: The normal distribution

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Radicals & Rational Exponents

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

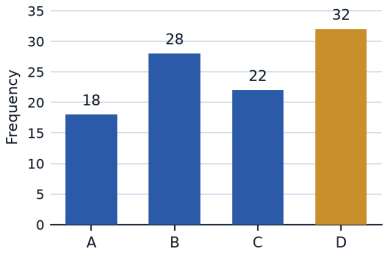
16 [Challenging] • Target: The normal distribution • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(3)$.

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17 [Challenging] • Target: The normal distribution • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



18 [Challenging] • Target: The normal distribution • Cross-Lesson: 9.01 Trigonometric Functions

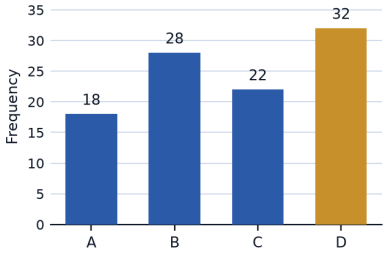
For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(4)$.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 1.01 Functions & Their Graphs

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

20 [Challenging] • Target: The normal distribution • Cross-Lesson: 10.01 Matrices & Systems

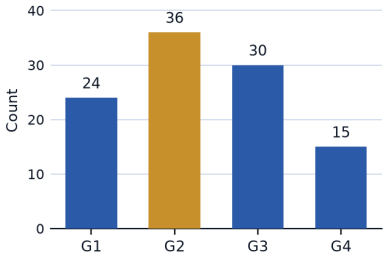
For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(5)$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(2)$.



a. 21

b. 27

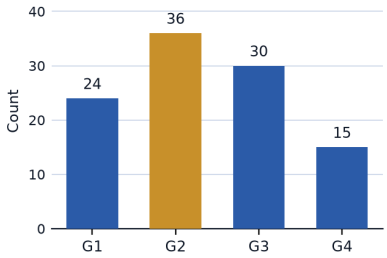
c. 12

d. 25

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22 [Foundational] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

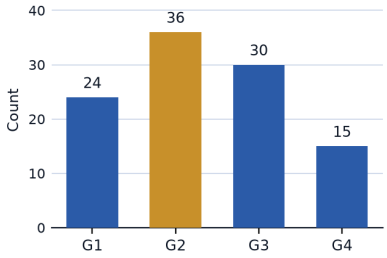
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

23 [Foundational] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(5)$.



a. 39

b. 45

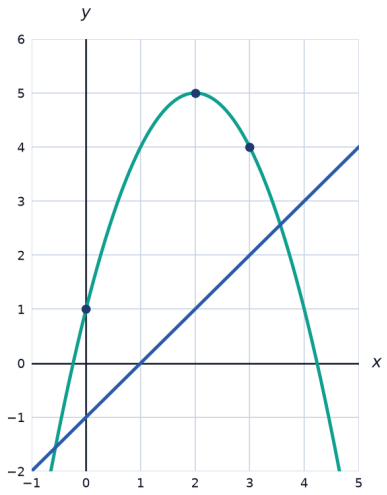
c. 30

d. 43

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24 [Foundational] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(2)$.



a. 7

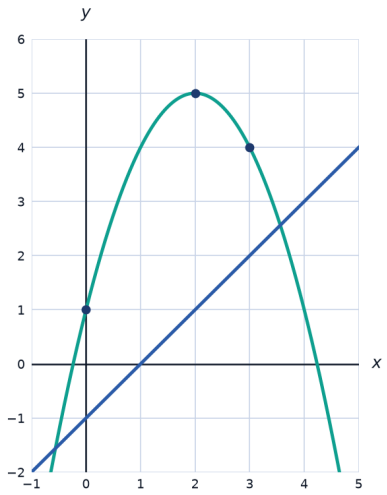
b. 9

c. 4

d. 11

25 [Foundational] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

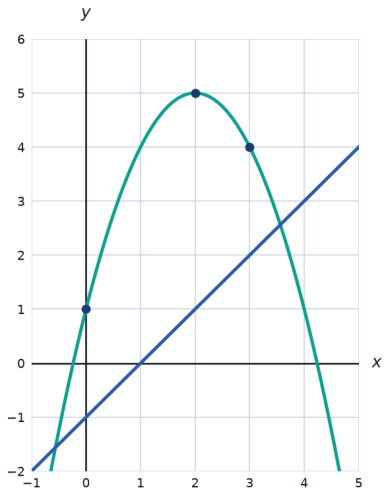
c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

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26 [Foundational] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(5)$.



a. 13

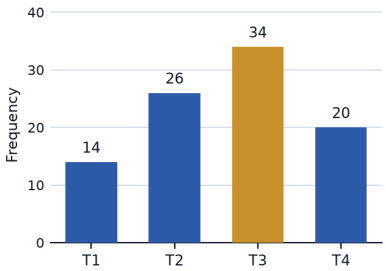
b. 15

c. 10

d. 17

27 [Foundational] • Target: Describing shapes of data distributions

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(2)$.



a. 10

b. 13

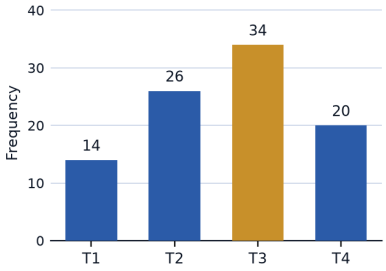
c. 6

d. 14

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28 [Foundational] • Target: Describing shapes of data distributions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Random sampling

A data set recorded for random sampling consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

31 [Intermediate] • Target: Random sampling

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(3)$.

a. 27

b. 33

c. 18

d. 31

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32 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

33 [Intermediate] • Target: Random sampling

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

34 [Intermediate] • Target: Estimating proportions from samples

A data set recorded for estimating proportions from samples consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

35 [Intermediate] • Target: Estimating proportions from samples

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

36 [Intermediate] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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37 [Intermediate] • Target: Estimating proportions from samples

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for describing shapes of data distributions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

39 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

40 [Challenging] • Target: Describing shapes of data distributions • Cross-Lesson: 10.01 Matrices & Systems

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

41 [Challenging] • Target: The normal distribution • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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42 [Challenging] • Target: Interpreting and comparing z-scores • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for interpreting and comparing z-scores consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

43 [Challenging] • Target: Interpreting and comparing z-scores • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 5x + 6$ in interpreting and comparing z-scores, evaluate $f(3)$.

a. 21

b. 26

c. 15

d. 25

44 [Challenging] • Target: Normal distribution probabilities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A data set recorded for normal distribution probabilities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

45 [Challenging] • Target: Normal distribution probabilities • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 6x + 7$ in normal distribution probabilities, evaluate $f(3)$.

a. 25

b. 31

c. 18

d. 29

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